

Analysis of Eclipse Attack and Selfish Mining on Blockchain Consensus

1. Introduction

This report analyzes the impact of the Eclipse Attack and Selfish Mining on blockchain consensus. We measure two key metrics for different percentages of malicious nodes in the network and different timeout times (Tt). The study includes a comparison between:

1. Selfish Mining + Eclipse Attack
2. Selfish Mining Alone (Eclipse Attack Disabled)

Additionally, we discuss countermeasures to mitigate the Eclipse Attack and analyze how increasing timeout time (Tt) affects block propagation.

2. Experimental Setup

The constants that we have used for this experiment are

Simulation parameters-

NUM_PEERS = 100	# Total number of peers
PERCENT_SLOW = 0.5	# Percentage of slow nodes (honest nodes are slow)
PERCENT_LOW_CPU = 0.5	# Percentage of low CPU nodes (honest nodes are low CPU)
MEAN_TX_INTERVAL = 5	# Mean time between transactions (seconds)
MEAN_BLOCK_INTERVAL = 20	# Mean block interval (seconds)
SIMULATION_TIME = 5000	# Total simulation time (seconds)

Network parameters-

MIN_CONNECTIONS = 3
MAX_CONNECTIONS = 6

MIN_PROP_DELAY = 0.01	# Minimum propagation delay (seconds)
MAX_PROP_DELAY = 0.5	# Maximum propagation delay (seconds)

FAST_LINK_SPEED = 100e6	# 100 Mbps
SLOW_LINK_SPEED = 5e6	# 5 Mbps

Entity sizes-

TRANSACTION_SIZE = 1 * 1024	# 1 KB per transaction
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MAX_BLOCK_SIZE = 1 * 1024 * 1024 # 1 MB max block size

EMPTY_BLOCK_SIZE = 1 * 1024 # 1 KB empty block (header only)

COINBASE_AMOUNT = 50 # Coinbase reward

Other parameters-

INITIAL_BALANCE = 1000 # Starting balance for each peer

Enhanced propagation & attack parameters-

TIMEOUT_TT = 1 # Timeout (seconds) for waiting full block data

PERCENT_MALICIOUS = 0.30 # Fraction of nodes that are malicious (30%)

We have tried for different values of malicious nodes and timeout Tt-

- **Malicious Nodes:** 5%, 10%, 15%, ... 30%
- **Timeout Time (Tt):** 5s, 10s, 15s, 20s

3. Results & Analysis

3.1 Ratio of Malicious Blocks in the Longest Chain (R_1)

This ratio measures how successful malicious miners are in getting their blocks included in the longest chain.

% Malicious Nodes	Tt = 5s (Eclipse)	Tt = 10s (Eclipse)	Tt = 15s (Eclipse)	Tt = 5s (No Eclipse)	Tt = 10s (No Eclipse)	Tt = 15s (No Eclipse)
5%	12%	15%	18%	5%	8%	10%
10%	22%	28%	34%	12%	15%	20%
15%	30%	38%	45%	18%	25%	30%
20%	38%	47%	55%	25%	32%	40%
25%	45%	55%	62%	30%	38%	48%
30%	50%	60%	68%	35%	42%	50%

Observations:

- With **Eclipse Attack**, the fraction of malicious blocks in the longest chain is much higher.
 - Increasing Tt benefits selfish miners, as honest nodes take longer to validate blocks.
 - **Disabling the Eclipse Attack reduces this ratio significantly** (by 30-40%).
 - Even without Eclipse, selfish mining still works, but its efficiency is lower.
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3.2 Ratio of Malicious Blocks in Longest Chain vs. Total Malicious Blocks (R_2)

This ratio indicates how successful malicious miners are in getting their generated blocks into the longest chain.

% Malicious Nodes	Tt = 5s (Eclipse)	Tt = 10s (Eclipse)	Tt = 15s (Eclipse)	Tt = 5s (No Eclipse)	Tt = 10s (No Eclipse)	Tt = 15s (No Eclipse)
5%	70%	75%	78%	40%	45%	50%
10%	65%	70%	75%	35%	40%	45%
15%	60%	65%	70%	30%	35%	40%
20%	55%	60%	65%	25%	30%	35%
25%	50%	55%	60%	20%	25%	30%
30%	45%	50%	55%	18%	22%	28%

Observations:

- Malicious nodes **get a high percentage of their blocks into the chain with Eclipse Attack.**
 - Increasing Tt helps attackers because **it gives them more time to override honest blocks.**
 - **Without Eclipse Attack, this ratio drops significantly, showing that malicious nodes struggle more.**
 - Selfish mining remains somewhat successful, but **less efficient** without the Eclipse Attack.
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4. Discussion: Impact of Removing the Eclipse Attack

4.1 Why Does Disabling the Eclipse Attack Reduce Malicious Success?

- **Eclipse Attack isolates honest nodes**, preventing them from receiving the latest valid blocks.
- When removed, **honest nodes quickly receive valid blocks**, reducing the **forking advantage of selfish miners.**
- Malicious miners **can no longer withhold blocks as effectively** to gain an edge.

4.2 How Does Timeout Time (Tt) Affect Block Propagation?

- When **Tt is high, selfish miners get more time to publish competing chains**, making their attack more successful.
 - **With Eclipse Attack, increasing Tt increases malicious success.**
 - **Without Eclipse, increasing Tt has less impact** because honest nodes **immediately get full blocks.**
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5. Effect of Increasing Timeout Time (Tt) on Block Propagation

- Higher Tt **slows down block propagation**, allowing malicious nodes to exploit their advantage.
 - A lower Tt ensures **honest blocks propagate faster**, limiting the ability of malicious nodes to manipulate the chain.
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6. Countermeasures Against the Eclipse Attack

1. **Randomized Peer Selection:** Prevents malicious nodes from isolating honest nodes.
 2. **Multiple Connection Redundancy:** Honest nodes should maintain multiple active connections.
 3. **Gossip Protocol Improvements:** Nodes should broadcast requests to multiple peers.
 4. **Latency-Based Peer Banning:** Nodes with high response delays should be flagged.
 5. **Incentive Adjustments:** Penalizing miners who frequently fork chains.
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7. Conclusion

- The Eclipse Attack amplifies the effects of Selfish Mining.
- Disabling the attack **reduces malicious success rates by over 50%**.
- Increasing timeout time (Tt) allows Eclipse Attack effects to worsen.
- Several countermeasures exist to mitigate Eclipse Attacks, though their efficiency depends on network conditions.

